

ACM RECSYS 2026 · RECSYS CHALLENGE WORKSHOP

# State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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Minneapolis · 2 October 2026 · Music-CRS track, team npatta01

47,071

tracks in the catalog

11

ways to search

1

learned ranker

A decorative bar chart at the bottom of the slide, consisting of numerous vertical bars of varying heights in a dark gray color.



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SIX PARTS

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# 1 The problem

finding music by talking





# Finding music by talking

THE PROBLEM

I'm trying to remember a song that feels like a quiet, personal story...



Neko Case — “Calling Cards”

Not quite the one... it definitely had a female singer.



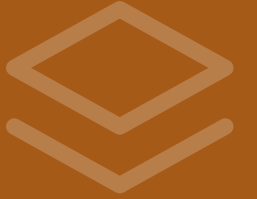
Neko Case — “Man”

No, that's not it either. Much more subdued... almost a cappella.



Your system answers here: **20 tracks** and a **short reply**

A real conversation from the challenge's final test set.



2

# Where the data comes from

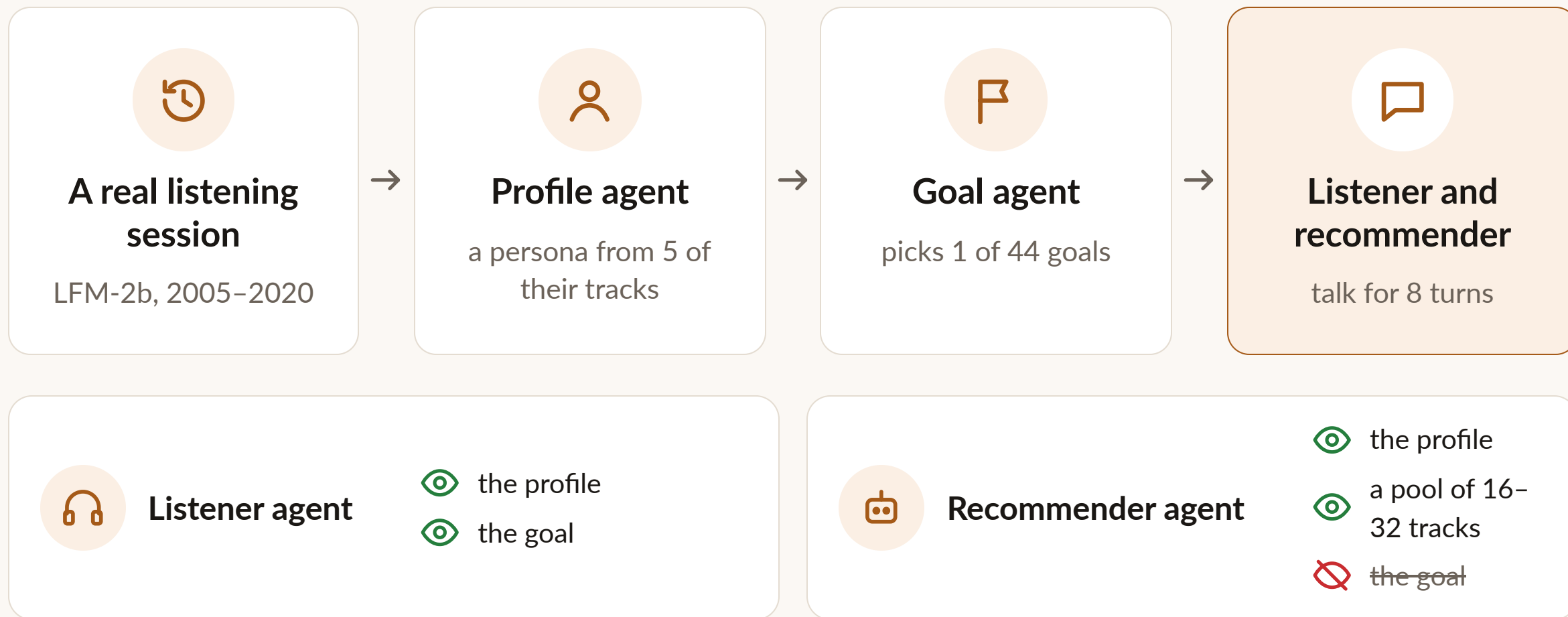
how the organisers built the conversations





# How the conversations were made

THE DATA



All four agents are **Gemini 2.5 Flash**. The recommender's pick **is the answer key**.

Source: TalkPlayData 2 (Doh et al., 2025), §3



# What you get

THE DATA



**15,199**

**training  
conversations**

8 turns each, with the  
answer for every turn



**1,000**

**development  
conversations**

with answers — for  
checking yourself



**2 × 80**

**blind test  
conversations**

final turn only; scored  
on the leaderboard



**47,071**

**tracks in the  
catalog**

title, artist, tags, plus  
text, audio and cover  
embeddings



# 3 How it is scored

one right track per turn, plus a judged reply





# The final score

SCORING

50%

right track · nDCG@20

30%

reply quality · LLM judge, 1–5

10%

catalog  
diversity

10%

reply wording

## LLM JUDGE

The organisers' LLM reads the conversation and your reply, and rates it 1–5.

## CATALOG DIVERSITY

How many **different** tracks you recommend across all turns, as a share of the catalog.

## REPLY WORDING

Lexical diversity: how varied the words in your replies are.



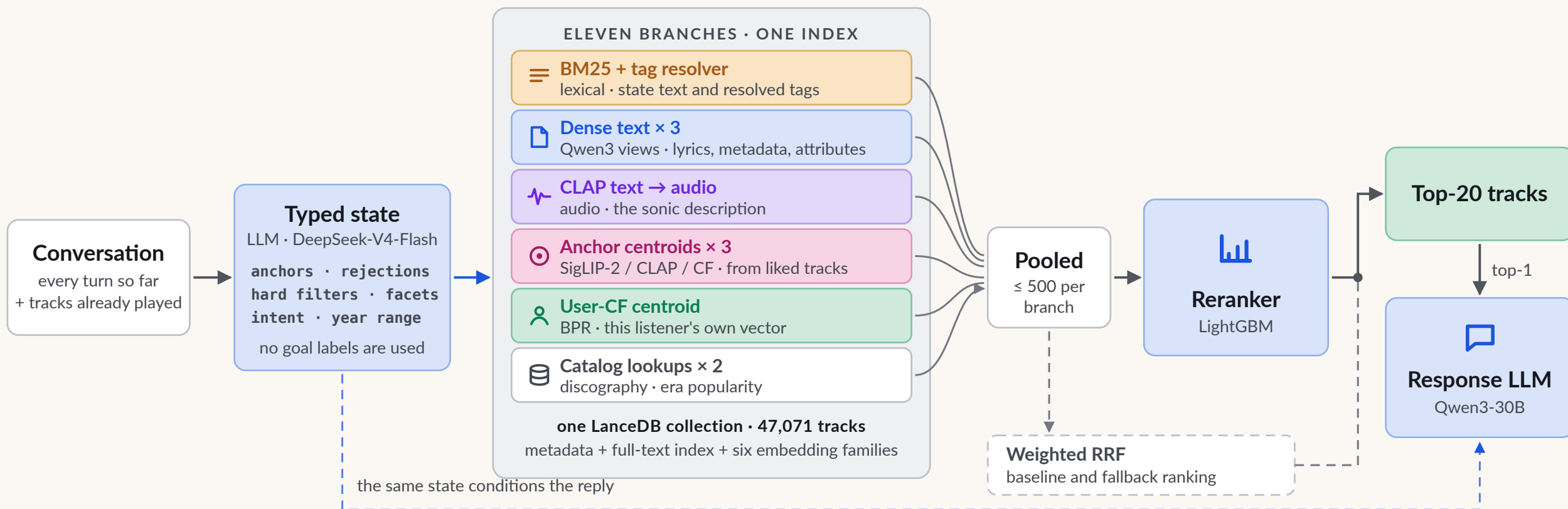
# 4 Our approach

request understanding, retrieval, reranking, response generation



# ≡ One state, eleven branches, one learned ranker

APPROACH



Organisers supplied the audio (CLAP), cover-art (SigLIP-2), listening-data (BPR) and small text (Qwen3-0.6B) embeddings; we added Qwen3-8B text embeddings and our own bi-encoder.

4.1

Request understanding

4.2

Retrieval

4.3

Reranking

4.4

Response generation

# 4.1

## Request understanding

reading the conversation into a state

# Request understanding

APPROACH

WE HAD JUST PLAYED



Ryan Adams & The Cardinals — “Let Us Down Easy”

DeepSeek  
V4-Flash



LISTENER

...more tracks with that kind of emotional depth and acoustic focus, maybe branching out to other singer-songwriter or alt-country artists?

CONVERSATION STATE

REQUEST TYPE    **new artist**

AVOID                **Ryan Adams & The Cardinals**

MOOD                **introspective, emotional depth**

SOUND               **acoustic-driven**

GENRE                **singer-songwriter, alt-country**

DeepSeek-V4-Flash reads **every turn so far** and writes a typed **state**: “other artists” after a Ryan Adams track becomes an artist to avoid, and the mood carries over.



# What the state can hold

APPROACH

## THE REQUEST

- request type
- intent mode
- routing flags

## WHAT THEY LIKE

- anchors and pinned tracks
- per-track feedback
- mentioned entities

## WHAT THEY RULED OUT

- rejected tracks
- rejected artists
- hard filters

## WHAT IT SOUNDS LIKE

- mood facets
- genre facets
- sonic facets

## WHAT IT IS ABOUT

- lyrical theme

## BOUNDS

- release-year window
- process constraints



# How the state drives retrieval

APPROACH



## The request

request type · intent mode · routing flags



Which branches fire, and how much each one weighs in the merge

routing weights

pivot skips anchors

gates discography



## What they like

liked tracks · named artists and tags



Search near liked tracks; match named entities

anchor centroids ×3

BM25 fields

artist discography



## What it sounds like · is about

mood · genre · sonic · lyrical theme



The text, audio and lyrics queries

dense text ×3

CLAP text→audio

lyrics view



## What they ruled out

rejected tracks and artists



Dropped from every branch before the ranker

hard drop

same-artist demote



## Bounds

release-year window · date filters



Mask out-of-range tracks; boost the era

candidate mask

era-popularity lookup

# Understanding a lookup

APPROACH

## LISTENER

Which song from Peter Doherty's *Grace/Wastelands* was mentioned as one of the best tracks of 2009?

## WE SERVED

 Peter Doherty — “Arcady”, from *Grace/Wastelands*

LLM



REQUEST TYPE

**exact album**

ARTIST

**Peter Doherty • must use**

ALBUM

**Grace/Wastelands • must use**

ALSO

**“best tracks of 2009”**

Two **hard** fields (artist, album) and one **soft** phrase. The request type switches on exact lookups.



# Understanding a visual request

APPROACH

LISTENER

An album with a really striking, abstract cover art... bold colors and shapes, almost like a modern painting.

LLM



|              |                        |
|--------------|------------------------|
| REQUEST TYPE | attribute search       |
| COVER ART    | striking, abstract     |
| COLOURS      | bold colors and shapes |
| STYLE        | like a modern painting |
| FLAG         | visual search          |

No artist, no title — only **descriptions**, plus a flag that turns on the cover-art search.

|                       |           |           |                     |
|-----------------------|-----------|-----------|---------------------|
| 4.1                   | 4.2       | 4.3       | 4.4                 |
| Request understanding | Retrieval | Reranking | Response generation |

# 4.2

## Retrieval

eleven searches over one index

# Retrieval — eleven branches

APPROACH



×1

## BM25 + tag resolver

keywords and resolved tags  
from the state



×3

## Dense text

Qwen3 embeddings of lyrics,  
metadata, attributes



×1

## CLAP text → audio

the sonic description, matched  
to audio



×3

## Anchor centroids

near liked tracks: cover art,  
audio, listening



×1

## User-CF centroid

this listener's own BPR vector



×2

## Catalog lookups

the artist's discography; era  
popularity

---

Each branch returns its **top 500**; the ranker sees their union.



# One index

## APPROACH

| WHAT IT DESCRIBES  | MODEL                       | SOURCE                   |
|--------------------|-----------------------------|--------------------------|
| Lyrics             | Qwen3-Embedding-0.6B        | organisers               |
| Metadata text      | Qwen3-Embedding-8B          | <a href="#">we built</a> |
| Attributes text    | Qwen3-Embedding-8B          | <a href="#">we built</a> |
| Audio              | LAION-CLAP                  | organisers               |
| Cover art          | SigLIP-2                    | organisers               |
| Listening data     | BPR collaborative filtering | organisers               |
| Conversation match | our fine-tuned bi-encoder   | <a href="#">we built</a> |

### ONE LANCEDB TABLE

Metadata, a full-text index and every embedding for all 47,071 tracks — each branch is a different query against the same table.

### PROVIDED VS BUILT

The organisers supplied four kinds of embedding. We added larger text embeddings and our own bi-encoder.

4.1  
Request understanding

4.2  
Retrieval

4.3  
**Reranking**

4.4  
Response generation

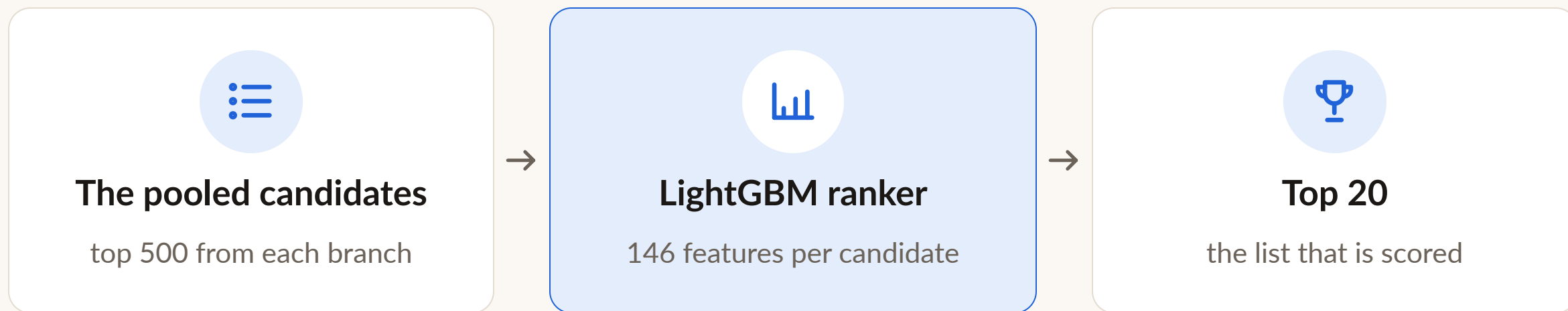
# 4.3

## Reranking

one learned model picks the 20

# Reranking

APPROACH





# What the ranker leans on

APPROACH

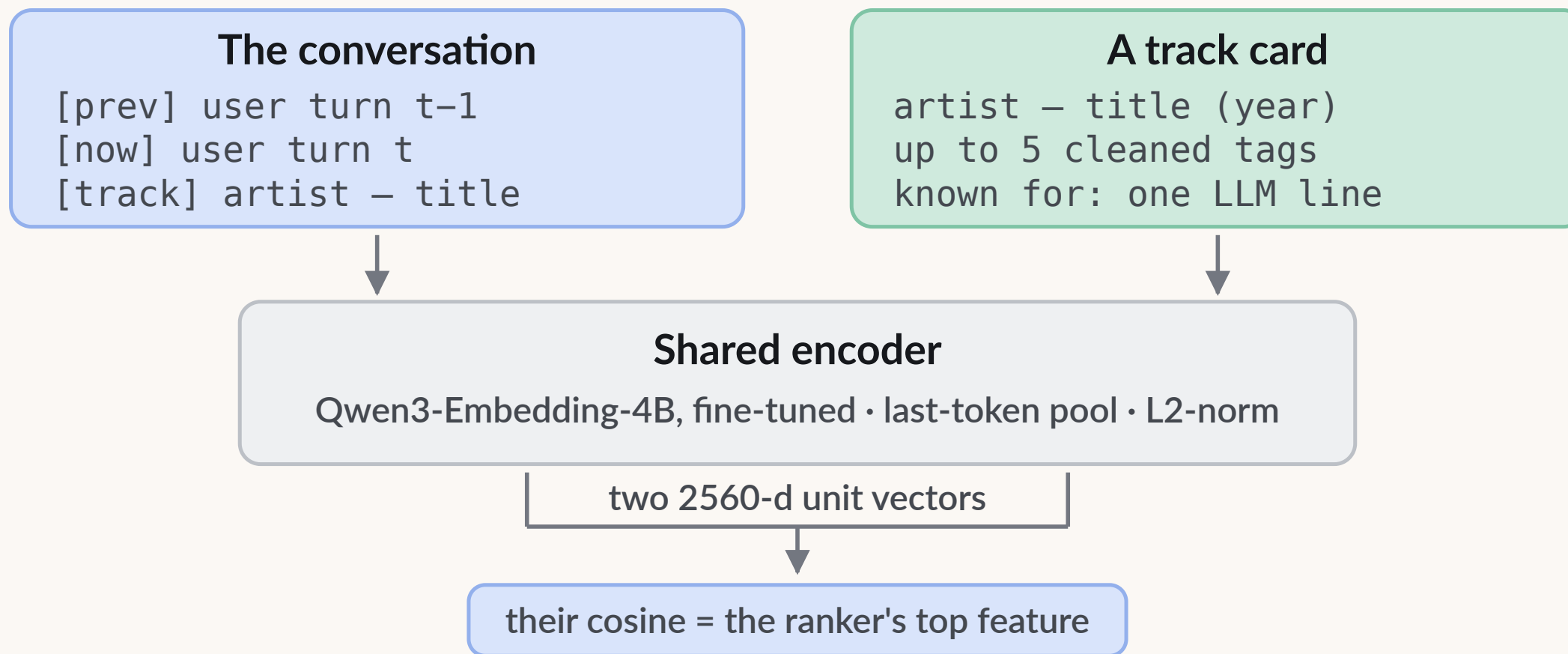


Share of the ranker's split gain, by feature family. One signal — the bi-encoder cosine — carries **59%**; constraints and rejections carry **0.7%**.



# The bi-encoder behind that top feature

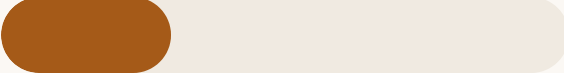
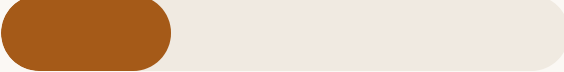
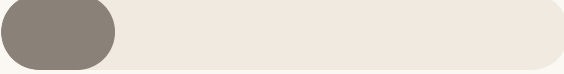
APPROACH





# Down-weighting contradicted turns

APPROACH

|                              |                                                                                      |       |
|------------------------------|--------------------------------------------------------------------------------------|-------|
| Every turn starts at         |    | 1.0   |
| Next turn rejects the artist |    | × 0.6 |
| Next turn rejects the track  |    | × 0.3 |
| Goal note: did not help      |    | × 0.3 |
| Never below                  |  | 0.2   |

## WHY

The label is whatever played next. When the **next** turn says it was wrong, that label counts for less in training.

Chosen by hand, never tuned. Goal notes shape these weights but are never a ranker input.

4.1  
Request understanding

4.2  
Retrieval

4.3  
Reranking

4.4  
Response generation

# 4.4

## Response generation

one short reply about the top track



# Response generation

APPROACH

## INPUTS

### Conversation

"...it feels a bit too serious. I'm looking for something more uplifting and positive..."

### Latest state

Mood **uplifting, positive**

Avoid **serious**

### Top-ranked track

Track **Röyksopp – "Happy Up Here"**

Tags **cheerful, hopeful, electronic...**



**Qwen3-30B**

temperature 0 • one  
pass

1-2 sentences about  
the top track only



## OUR REPLY

"Perfect – 'Happy Up Here' by Röyksopp is exactly the kind of uplifting, cheerful electronic track you're looking for. With its bright synths and hopeful groove, it's designed to lift your spirits and make you smile."



# How the reply is written

APPROACH

## Latest state

|       |                                   |
|-------|-----------------------------------|
| Genre | <b>electronic</b>                 |
| Mood  | <b>uplifting, positive, happy</b> |
| Avoid | <b>serious</b>                    |

## Top track

|       |                                                         |
|-------|---------------------------------------------------------|
| Track | <b>Röyksopp – “Happy Up Here”</b>                       |
| Tags  | <b>cheerful, hopeful,<br/>experimental, Pop, 00s...</b> |



“Write 1–2 concise sentences about only the selected track. Prioritize the latest user request and extracted state over older conversation history. If the track is reasonably aligned, explain the fit with one specific supported reason. If it clearly conflicts with an explicit avoid/new-artist constraint, do not oversell it or blame the system; briefly frame the limitation and the closest supported reason.”

Qwen3-30B-A3B-Instruct

temperature 0

one pass

output: the reply on the previous slide



# 5 How it scored

the official results

## ✓ How it scored

RESULTS

| OFFICIAL CODABENCH SPLIT    | NDCG@20 | CAT.   | LEX.   | JUDGE | COMPOSITE |
|-----------------------------|---------|--------|--------|-------|-----------|
| Blind-A · development phase | 0.4380  | 0.0313 | 0.8028 | 4.70  | 0.5799    |
| Blind-B · final             | 0.2537  | 0.0315 | 0.7862 | 3.30  | 0.3811    |

NDCG@20

**-42%**

0.4380 → 0.2537

JUDGE (1-5)

**-1.40**

4.70 → 3.30

DIVERSITY TERMS

**flat**

catalog and reply-wording  
diversity barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.



## Where we placed

RESULTS

#1 · hallucinated



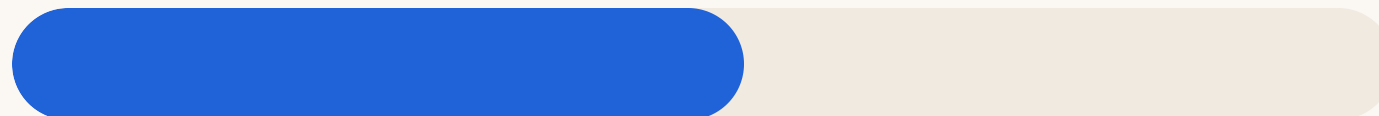
0.689

Median of 41 teams



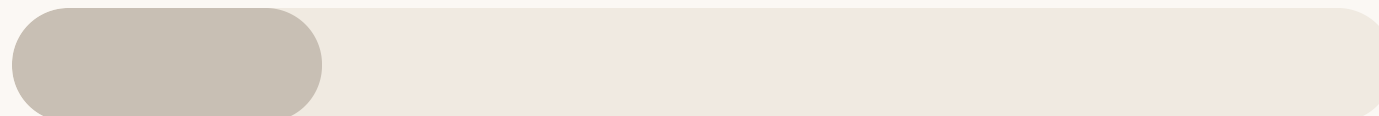
0.474

#29 · npatta01 (us)



0.381

Organisers' BM25 baseline



0.162

29th: well above the baseline, below the median.

Final (Blind-B) composite score. Source: official Music-CRS results table, [nlp4musa.github.io/music-crs-challenge/results.html](https://nlp4musa.github.io/music-crs-challenge/results.html) — 41 teams listed; verification in progress.



# 6 What is not ideal

what went wrong, one finding per slide



# What is not ideal

## LESSONS

We trained and validated the ranker on the **same dev turns**.

THE LISTENER ASKED FOR

We extracted more than we used

*the Kamelot track from **Silverthorn***



OUR TOP PICK

Kamelot — Fallen Star, from **Haven**

We extracted “**Silverthorn**” but never filtered on it

We used no data from outside the catalog

*exactly **126.70 bpm**, in G minor*



the catalog has no tempo or key, and we added no outside source

The reply did not check our pick against the request

*“**Watercolors**” by Pat Metheny*



Metheny — Alfie, with the reply “Got it — here’s ‘Alfie’...”

“**Watercolors**” is not in the catalog



# Label example — “My Body Is a Cage”

WHAT IS NOT IDEAL

RECOMMENDER AGENT PLAYED



Arctic Monkeys — “Brianstorm”

LISTENER, NEXT TURN

This is still Arctic Monkeys. I keep asking for ‘My Body Is a Cage’ by **Arcade Fire**... That’s the only song I want to hear right now.

ANSWER KEY FOR THAT TURN



Arctic Monkeys — “This House is a Circus”

DOES NOT MOVE TOWARD GOAL

✓ Our judges agree: wrong — repeats the artist

The answer key is whatever the simulated recommender picked from its small pool — here, **Arctic Monkeys** again.

# Inconsistent labels — “from a different artist”

WHAT IS NOT IDEAL

RECOMMENDER AGENT PLAYED



Bonobo — “Cirrus”

LISTENER, NEXT TURN

...something with a similar chill, electronic vibe, maybe with a subtle beat, but **from a different artist?**

ANSWER KEY FOR THAT TURN



Bonobo — “Jets”

MOVES TOWARD GOAL

✗ Our judges: wrong — same artist

NEXT TURN

“Jets” is cool, but I was actually hoping for something **from a different artist this time...**

ANSWER KEY



Bonobo — “Pieces”

MOVES TOWARD GOAL

✗ Our judges: wrong — same artist

Twice asked to leave Bonobo; both answer keys are Bonobo, **marked helpful**.

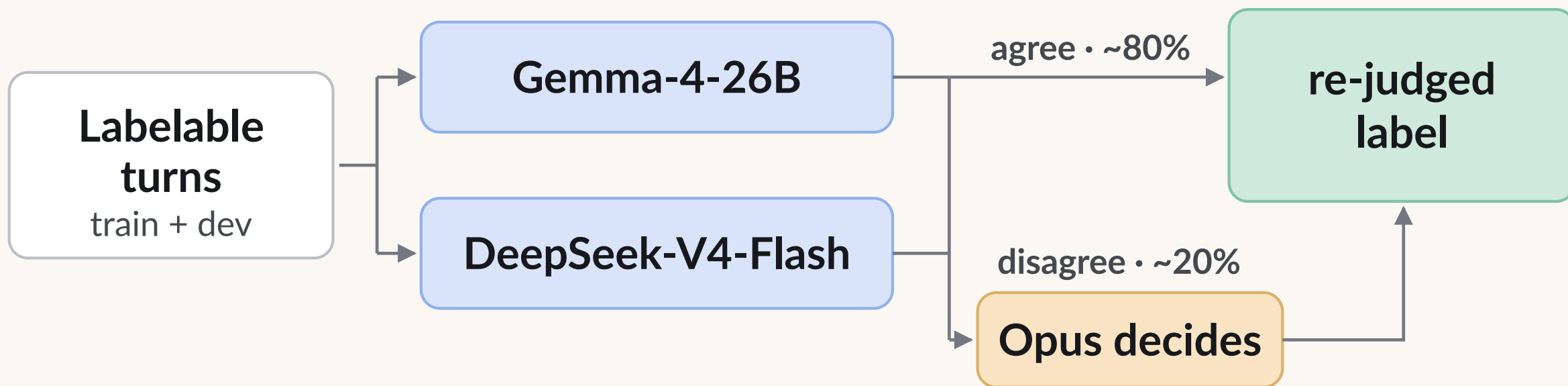


# How the LLM relabelling worked

ANALYSIS

The answer keys come from the simulated recommender's picks. To see how often they actually fit the request, we had **two LLM judges re-score every labelable turn**.

Both judges score the same two axes: was a different artist asked for, and does the track fit the request?



Agreement split measured on the training turns.



# What the relabelling found

ANALYSIS

## 17%

of training turns keep the artist the listener had just asked to leave

18,222 turns — both judges or the arbiter marked the label as artist anchoring

## 113,393

turns re-judged

every labelable train and dev turn

---

Retrained on these labels, the ranker scored lower **against the original labels** — a test that could never favour the fix — so we kept the originals.



# Why we trained on the dev set

LESSONS



**State extraction: 1 paid LLM call per turn**

to read each turn into a state



**Too costly to extract state for 121.6k training turns**

so no ranker features there



**Trained on the 8,000 dev turns**

the same turns we checked it on

Dev, turns it trained on



0.384

Blind-B, official



0.254

nDCG@20. The dev row is an earlier model version, shown because it is the number we selected on. Source: paper §4.3, §5.3.



# Extracted, but the ranker did not act on it

WENT WRONG

|                                                                               |                                                                               |                                                                                  |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <p>"I gotta break away from <b>Deltron</b> and <b>Del</b> completely"</p>     | <div><div>✓</div><div>STATE EXTRACTED<br/>avoid Deltron 3030, Del</div></div> | <div><div>✗</div><div>OUR TOP 20<br/>10 of 20 by them — from rank 6</div></div>  |
| <p>"branching out to <b>other</b> singer-songwriter artists"</p>              | <div><div>✓</div><div>STATE EXTRACTED<br/>avoid Ryan Adams</div></div>        | <div><div>✗</div><div>OUR TOP 20<br/>10 of 20 by Ryan Adams — rank 1</div></div> |
| <p>"still focusing on <b>Juanes</b>... haven't found the depth I'm after"</p> | <div><div>✓</div><div>STATE EXTRACTED<br/>rejected Juanes</div></div>         | <div><div>✗</div><div>OUR TOP 20<br/>7 of 20 by Juanes — rank 1</div></div>      |



# No data from outside the catalog

WHAT IS NOT IDEAL

## A tempo and a key

*“exactly 126.70 bpm, in G minor”*



the catalog has no tempo or key field  
**no branch can match on either**

## A soundtrack

*“a specific 80s synth-pop song from the **Breaking Bad** soundtrack”*



we served Soft Cell — “Tainted Love”  
**no film or TV links in the catalog**

---

The state was right both times. The gap is **knowledge, not understanding**.



# Gaps in our response generation

WHAT IS NOT IDEAL

The track is not in the catalog, so retrieval could not return it — the reply presents our substitute as if it were the request

1

*"I wanted 'Watercolors' by Pat Metheny specifically."*

"Got it — here's 'Alfie' by Pat Metheny..."

NOT FLAGGED

Our pick broke the listener's constraint — the reply recommends it anyway

2

*"...something from a **different artist** this time"*

"...while this track is by him, it's from a different era and band..." — Ryan Adams again

NOT FLAGGED

---

One pass, no check: the reply could flag **neither unavailable tracks nor unmet constraints**. The prompt was chosen on Blind-A (judge 4.70 there, 3.30 on Blind-B).



# Everything is released

THANK YOU



Paper, poster, slides, code, models, data

[npatta01.github.io/music-crs-2026](https://npatta01.github.io/music-crs-2026)

# → Appendix

TYPE A NUMBER + ENTER · OR G TO SEARCH

- |                    |                            |                    |                                        |                    |                                                      |
|--------------------|----------------------------|--------------------|----------------------------------------|--------------------|------------------------------------------------------|
| <a href="#">1</a>  | Where our points came from | <a href="#">2</a>  | The task                               | <a href="#">3</a>  | Glossary                                             |
| <a href="#">4</a>  | State extraction           | <a href="#">5</a>  | Six request types, six shapes of state | <a href="#">6</a>  | Routing flags decide which branches fire             |
| <a href="#">7</a>  | Entity resolution          | <a href="#">8</a>  | The eleven branches                    | <a href="#">9</a>  | Fusion and fallback                                  |
| <a href="#">10</a> | b1 training                | <a href="#">11</a> | Development-split results              | <a href="#">12</a> | Failure 3 — the better track was already in the pool |
| <a href="#">13</a> | Label-free audit           | <a href="#">14</a> | Limits of our own analysis             | <a href="#">15</a> | What we would do differently                         |
| <a href="#">16</a> | Related work               | <a href="#">17</a> | Reproducing the run                    | <a href="#">18</a> | The target often ignores the request                 |

# Where our points came from

RESULTS

$$0.50 \cdot \text{nDCG@20} + 0.10 \cdot \text{cat.div} + 0.10 \cdot \text{lex.div} + 0.30 \cdot (\text{judge} - 1)/4$$

## OUR BLIND-B DECOMPOSITION

|                   |        |        |
|-------------------|--------|--------|
| 0.50 × nDCG@20    | 0.2537 | 0.1269 |
| 0.10 × cat. div.  | 0.0315 | 0.0032 |
| 0.10 × lex. div.  | 0.7862 | 0.0786 |
| 0.30 × (3.30−1)/4 |        | 0.1725 |
| Composite         |        | 0.3811 |

## WHERE THE SCORE CAME FROM

The **judge term** out-earned **nDCG**, and catalog diversity returned **0.003** of the 0.10 on offer. We optimised almost entirely for relevance and left the diversity terms alone.

## EVERY ASSISTANT TURN

- Retrieve **20 tracks** from a shared **47,071-track** catalog
- Write a short reply presenting the top recommendation

## PER-TRACK METADATA

Title, artist, album, a median of **17** genre/mood tags, popularity, release date, plus precomputed text, audio, cover-art and collaborative-filtering embeddings.

## SPLITS

|                   |                                  |
|-------------------|----------------------------------|
| Train             | 15,199 sessions (~121.6k turns)  |
| Development       | 1,000 sessions × 8 turns = 8,000 |
| Blind-A / Blind-B | 80 sessions each, final turns    |

Blind-B additionally withholds the conversation-goal annotations present in training data.

## BUILT ON

TalkPlayData 2. The conversations are **LLM-generated**, not observed listening sessions — which matters later.



# Glossary

BACKGROUND

## Typed state

Structured fields an LLM reads off the conversation each turn: artists to anchor on, rejected tracks, filters, mood.

## Dense embedding

A vector from a neural model; texts with similar meaning land close together.

## SigLIP-2

The same idea for text and images — here, album covers.

## Centroid search

Average the vectors of tracks the listener liked, then search near that average.

## LambdaMART · LightGBM

Gradient-boosted decision trees trained to order a list; LightGBM is the library.

## LLM judge

An LLM that rates the reply 1–5; the organisers run it.

## BM25

A classic keyword-search score over text fields.

## CLAP

A model that puts text and audio in one space, so a description can find a sound.

## Collaborative filtering (BPR)

Vectors learned from who listened to what; BPR is the training method.

## RRF

Reciprocal-rank fusion: merge ranked lists by summing  $1 / (k + \text{rank})$  per track.

## Bi-encoder

One model encodes the conversation and the track separately; relevance is the cosine of the two vectors.

## In-sample vs out-of-fold

Scoring on turns the model trained on, versus on held-out folds. In-sample overstates.



# State extraction

APPROACH

## THE EXTRACTOR

DeepSeek-V4-Flash (build 0423), schema-constrained, given the dialog so far and the IDs of tracks already played.

## DELIBERATELY BLIND

No conversation goals, goal progress, or user profiles reach the extractor. The state is whatever an LLM can read off the conversation.

## WHY THIS MODEL

Cheaper extractors under-filled the optional fields — rejections and facets especially — so every final run uses this one. Per-turn extraction is a paid API call, which is what kept us off the training split.

## PROMPT VERSION

ConversationStateV1, projected to V0Plus for the downstream contract.



# Six request types, six shapes of state

APPROACH

| REQUEST TYPE                  | A REAL TURN                                                         | WHAT THE EXTRACTOR PULLED OUT                                                                    |
|-------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <code>exact_track</code>      | "I wanted <i>Watercolors</i> by Pat Metheny specifically."          | track = <i>Watercolors</i> <b>must_use</b> · artist = Pat Metheny <b>must_use</b>                |
| <code>exact_album</code>      | "Which song from Peter Doherty's <i>Grace/Wastelands...</i> "       | artist + album <b>must_use</b> · facet "best tracks of 2009"                                     |
| <code>same_artist</code>      | "Got another track known for his guitar work?"                      | artist <b>must_use</b> · prior track <b>do_not_use</b> · facets: guitar work, melodic complexity |
| <code>new_artist</code>       | "...branching out to other singer-songwriter artists."              | artist = Ryan Adams & The Cardinals <b>do_not_use</b> · facets: acoustic, introspective          |
| <code>attribute_search</code> | "...a really striking, abstract cover art, bold colors and shapes." | three visual facets · routing flag <b>image_or_visual_search</b>                                 |
| <code>hidden_target</code>    | "A Lil Wayne song from an experimental phase."                      | artist <b>must_use</b> · facet "experimental" · flag <b>hidden_target_search</b>                 |

The extractor read all six of these correctly. Across the whole Blind-B run only **3 of 80 turns** were diagnosed as a state-extraction miss — reading the conversation was almost never the problem.



# Routing flags decide which branches fire

APPROACH

| FLAG                   | OPENS                                    |
|------------------------|------------------------------------------|
| exact_entity_probe     | catalog lookups — discography, album     |
| lyric_search           | the Qwen3 lyrics view                    |
| feature_articulation   | CLAP text→audio on the sonic description |
| image_or_visual_search | the gated SigLIP-2 cover-art branch      |
| hidden_target_search   | widens the pool; suppresses exact pins   |

## A REAL VISUAL REQUEST

“I’m looking for an album with a really striking, abstract cover art... bold colors and shapes, almost like a modern painting.”

→ three visual facets, `image_or_visual_search = true`

## AND THE HONEST PART

The reranker carries **no features** for that branch. Even when the flag opens it, the ranker cannot weigh what it returns.



# Grounding names to the catalog

APPROACH

## NAMES → IDS

Fuzzy string matching of artist and track names against the catalog name lists, keeping matches above **80/100**. Downstream stages only ever see identifiers.

## FREE TEXT → TAGS

- 1 Exact match against the tag vocabulary
- 2 Alias match
- 3 Substring match
- 4 Embedding nearest neighbour, cosine  $\geq 0.6$ , top 3

Used by the BM25 branch. Outside it, tag matching falls back to exact strings — part of why so little of the state became executable.

# The eleven retrieval branches

APPROACH

| BRANCH               | SPACE            | QUERY                       |
|----------------------|------------------|-----------------------------|
| BM25 (fielded)       | lexical          | state text + resolved tags  |
| Dense text × 3       | Qwen3 views      | state-built strings         |
| CLAP text→audio      | audio            | sonic description           |
| Anchor centroids × 3 | SigLIP-2/CLAP/CF | liked-track centroids †     |
| User-CF centroid     | CF-BPR           | the listener's own vector   |
| Lookups × 2          | catalog          | discography; era popularity |

## GATED TWELFTH

SigLIP-2 text→cover-art fires only on a visual request. The reranker carries no features for it.

## POOL DEPTH

Each branch truncated to its top 500 before the union reaches the ranker.

† Gated by intent mode: a pivot suppresses anchor centroids.

Appendix

B — back to the talk · A — index · number + Enter — jump



# Weighted RRF — baseline and fallback

APPROACH

## FUSION

Weighted reciprocal-rank fusion,  $k = 60$ , top-1000 across the branch pools.

## POST-FUSION ADJUSTMENTS

- Drop rejected artists and tracks
- Drop tracks already played
- Promote / demote on exact tag membership
- Year-range decay

## AS A BASELINE

**0.1492**

devset nDCG@20, no training

## STILL THE FALLBACK

If the learned ranker is unavailable, this order ships. It is the only part of the ranking stack with no trained component.



# How b1 was trained

APPROACH

$$L = -(1/B) \sum_i \log \left[ \exp(v_{q_i}^T v_{d_{i+}} / \tau) / \sum_{d \text{ in } D_i} \exp(v_{q_i}^T v_d / \tau) \right] \quad \tau = 1/20$$

## POSITIVES

**53,885**

(turn, track) pairs whose *next* turn's goal-progress says the recommendation moved the session forward

## NEGATIVES

**4 + batch**

four mined hard negatives per positive, plus every in-batch document

## RETRIEVAL VS RANKING

As a twelfth *retrieval* branch it added little to the union's coverage and was dropped. As a *ranking* feature it dominates the fit — out-of-fold nDCG@20 0.1970 → 0.2032.

Not isolated: which parts of the rendering matter, and whether the effect carries to the blind splits.



# Development split – what we steered on

RESULTS

| SYSTEM (HOW MEASURED)                       | NDCG@20       |
|---------------------------------------------|---------------|
| Weighted RRF fusion (no training)           | 0.1492        |
| LambdaMART, out-of-fold CV                  | 0.1970        |
| + b1_cos (deployed feature set)             | 0.2032        |
| Historical v10 LambdaMART, <b>in-sample</b> | 0.3844        |
| <b>Official Blind-B</b>                     | <b>0.2537</b> |

### DO NOT OVER-READ THE FIRST GAP

The fusion and learned-ranker configurations used **different candidate-pool depths**, so 0.149 vs 0.197–0.203 does not isolate the effect of learned re-ranking.

### THE IN-SAMPLE ROW

A historical goal-aware model without b1\_cos.  
Not directly comparable – shown because it is the number we were selecting on.



# Failure 3 — the better track was already in our pool

ANALYSIS

| WHAT THEY ASKED FOR                                            | WE SERVED AT RANK 1               | ALREADY IN OUR POOL                                    | POOL RANK |
|----------------------------------------------------------------|-----------------------------------|--------------------------------------------------------|-----------|
| nu-jazz, downtempo, female vocals —<br><i>not</i> Parov Stelar | Parov Stelar — Booty Swing        | St Germain — What You Think About...                   | 34        |
| new underground hip-hop, not Deltron                           | Dead Prez — Hip Hop               | Souls Of Mischief — Make Your Mind Up                  | 31        |
| hardcore, not classic hip-hop                                  | Dead Prez — Hip Hop               | Papa Roach — Getting Away With Murder                  | 49        |
| that carol, but not Beegie Adair again                         | Beegie Adair — The Christmas Song | Frank Sinatra — Have Yourself A Merry Little Christmas | 63        |

On **31 of 80 Blind-B turns** a non-violating, better-fitting candidate was sitting in our own pool below the top-20 we served. Retrieval found it; ranking buried it.



# Auditing the submitted run

ANALYSIS

## SINGLE-JUDGE QUALITY RATING

78%

of Blind-B turns rated weak-or-bad by one DeepSeek-V4-Flash judge

## READ AS

Diagnostic, not ground truth. Blind-B exposes no relevance labels, so only the validation failure is actually *measured*.

## DETERMINISTIC TRACE AUDIT — 43 DIAGNOSES OVER 32 TURNS

|                                    |    |
|------------------------------------|----|
| Heuristic alternative below top-20 | 31 |
| Under-applied constraint           | 8  |
| State-extraction miss              | 3  |
| Entity-resolution miss             | 1  |

Categories overlap. The alternative was chosen by positive-term metadata matches plus popularity — an audit heuristic, not a relevance judgment.



# Limits of our own analysis

ANALYSIS

- The relabeling was **never checked against human judgments**.
- The Blind-B quality audit used a **single** LLM judge.
- The trace audit was **heuristic**, not a relevance judgment.
- The conversations are **LLM-generated**, not observed listening sessions.
- Neither blind split exposes relevance labels — blind-set failure rates and relabeling rates are **diagnostic, not ground truth**.
- Development-split and leaderboard numbers are **separate measurements**.



# What we would do differently

CONCLUSION

- 1 **Hold out a clean split before touching anything.** Free, and it would have changed every selection decision we made.
- 2 **Make the state executable.** Rejections and constraints carried 0.7% of model gain — most of what the extractor understood never became a filter.
- 3 **Import knowledge the catalog lacks** — tempo, key, soundtrack and film/TV links — so requests naming them have something to match.
- 4 **Check the response against the state** before emitting it: flag unavailable tracks and unmet constraints instead of overselling a substitute.
- 5 **Decide what to do when the benchmark and your judgment disagree.** We still do not have a principled answer.

## → Related work — four established lines

A1

### CONVERSATIONAL STATE TRACKING

CRS systems commonly track dialog state explicitly and condition retrieval on it. We follow that, but compile the state with a **prompted LLM** instead of a trained tracker.

### MODALITY-SPECIFIC MUSIC ENCODERS

Contrastive language-audio pretraining (CLAP), vision-language over cover art (SigLIP-2), collaborative filtering (BPR), general text embeddings (Qwen3).

### RETRIEVE-THEN-RERANK AT CATALOG SCALE

Pool multiple sources with reciprocal-rank fusion; rank with LambdaMART.

### OUR CLAIM

What we report is **how that combination behaved** — not a new technique.



# Reproducing the run

A2

## FROZEN REPLAY – NO CREDENTIALS

Source the offline bundle's env and every LLM call is served from a curated cache. A miss hard-fails rather than silently spending money.

```
source .repro/scripts/activate_repro_env.sh
```

## LIVE RE-RUN

```
python run_experiment.py --backend modal --tid state_ranker_v10_lgbm_blindset_B --eval_dataset blindset_B
```

## ARTIFACTS

Models and data: [hf.co/datasets/Npatta01/music-crs-repro-2026](https://huggingface.co/datasets/Npatta01/music-crs-repro-2026)



# The target often ignores the request

ANALYSIS

LISTENER · JUST PLAYED BONOBO

...something with a similar chill, electronic vibe, but **from a different artist?**

ANSWER KEY



Bonobo — “Jets”

MOVES TOWARD GOAL

NEXT TURN

“Jets” is cool, but I was hoping for something **from a different artist this time...**

ANSWER KEY



Bonobo — “Pieces”

MOVES TOWARD GOAL

The answer key repeats the artist the listener just asked to leave. Our ranker **anchors** the same way — we think it learns it here.